

Input: training data $\{(x_i, y_i)\}_{i=1}^n$ and a differentiable loss $L(y, F(x))$

1. Start with the best constant: $F_0(x) = \arg \min_{\gamma} \sum_{i=1}^n L(y_i, \gamma)$

2. For $m = 1$ to M :

(a) pseudo-residuals: $r_{im} = - \left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \right]_{F=F_{m-1}}$

(b) fit a regression tree to the r_{im} : terminal regions R_{jm}

(c) leaf values: $\gamma_{jm} = \arg \min_{\gamma} \sum_{x_i \in R_{jm}} L(y_i, F_{m-1}(x_i) + \gamma)$

(d) update: $F_m(x) = F_{m-1}(x) + \sum_j \gamma_{jm} \mathbf{1}(x \in R_{jm})$

repeat
 M times

3. Output: $\hat{f}(x) = F_M(x)$